Kodak Once Had a Nuclear Machine in Its Basement

From: Popular Mechanics Mined: 2026-10-18 3 min read

For about 30 years, the film company Eastman Kodak kept an unusual machine in a secure basement in Rochester, New York. It was a nuclear device about the size of a refrigerator.

Kodak turned it on in 1975. The machine made a steady stream of **neutrons**. **Neutrons** are tiny particles found in the centers of **atoms**.

Kodak used the **neutrons** to test chemicals and find out what was in them. It also used them to make special pictures, a bit like X-rays. These pictures could show light materials, like water, that X-rays have trouble showing.

It might sound like a nuclear reactor, but it was built so that it could not keep a **chain reaction** going by itself.

The device ran until 2006 behind thick walls, under government security rules. Kodak and officials said it was tightly controlled. When it was taken apart, its **uranium** plates were carefully moved to a government facility. When people learned about it in 2012, many were alarmed. But the real story is a clever piece of **Cold War** engineering.

Word Treasure

neutrons -- Tiny particles found in the centers of atoms.

atoms -- The tiny building blocks that everything is made of.

chain reaction -- A process where each step starts the next one, so it keeps going.

uranium -- A heavy metal used in nuclear science.

Cold War -- A long time of tension between the United States and the Soviet Union.

Background you need

Everything is made of tiny atoms, and the centers of atoms hold even tinier particles called neutrons.

Some machines use neutrons to look inside things, a bit like X-rays.

Nuclear materials can be dangerous, so governments make strict safety rules for them.

Break it down

WHO: The film company Eastman Kodak
WHAT: It ran a nuclear machine that made neutrons for about 30 years
WHERE: In a secure basement in Rochester, New York
WHY: To test chemicals and make special pictures, a bit like X-rays

Why it matters

It shows how scientists can use powerful tools safely by following careful rules.

Test what you remember

1. Where did Kodak keep the machine?

(A) In a park
(B) In a secure basement in Rochester, New York
(C) On a boat
(D) In a school gym

2. About how big was the machine?

(A) About the size of a pencil
(B) About the size of a refrigerator
(C) About the size of a car tire
(D) About the size of a house

3. When did Kodak turn the machine on?

(A) 1975
(B) 1925
(C) 2012
(D) 2020

4. What did the machine make?

(A) Ice cream
(B) A steady stream of neutrons
(C) Photos of the moon
(D) Electricity for homes

5. Why was the machine different from a nuclear reactor?

(A) It used no electricity
(B) It was made of wood
(C) It could not keep a chain reaction going by itself
(D) It was too small to turn on

6. Where did the uranium plates go when the machine was taken apart?

(A) To another country
(B) To a government facility
(C) Into the trash
(D) To a museum

Answer key (try the quiz first!): 1: B 2: B 3: A 4: B 5: C 6: B

Questions to think about

1. Kodak and officials: Said the machine was tightly controlled.

MY THOUGHTS (at least 20 words):